

Awais Mohiuddin

Dr. Amrita George

Agentic AI

25 March 2026

Project Milestone 2: Multi-Agent Workflow

This milestone overviews a multi-agent hotel chatbot system that is designed to answer customer questions about policies, amenities, and services. This system uses a fixed pipeline where agents have specialized roles and operate sequentially. This allows for more structured and accurate responses for the customer.

Design

1. Classification Agent

Role: The Classification Agent is responsible for reading the user query and identifying the type of request the user is making based on input. It will take the hotel related question and separate it into categories such as check-in, check-out, amenities, parking, etc.

Input: User Query

Output: Category Label

Communication: The Classification Agent passes on the category label to the retrieval agent so it can guide its search for policy information.

2. Retrieval Agent

Role: The Retrieval Agent uses the category label along with the original user query to search the hotel policy knowledge base. It uses the label and the query to find any

relevant policy information that can fulfill the user's request. This agent is responsible for making sure the chatbot stays within policy guidelines and does not begin using general knowledge or start hallucinating. It only pulls information from the knowledge base data source. In the case that there is a failure, or the agent is unable to find relevant information, it will return an error saying, "No Relevant Policy Found".

Input: Category Label, User Query, Hotel Policy Knowledge Base

Output: Relevant Hotel policy information or "No_Relevant_Policy_Found"

Communication: The Retrieval Agent passes on the relevant hotel policy information to the response agent so it can formulate a response.

3. Response Agent

Role: The Response Agent uses the hotel policy information to generate a response that can be presented to the user. Its purpose is to provide a warm and professional response as if it were a hotel concierge. The Response Agent has constraints in place so that it only bases responses on relevant hotel policies. This helps keep the responses specific to the hotel and avoids the bot rely on general information.

Input: Retrieved Hotel Policy Text, User Query

Output: Draft Response

Communication: The Response Agent passes on the draft response to the critic agent so it can undergo review.

4. Critic Agent

Role: The Critic Agent reviews the draft response that was generated by the Response Agent. The purpose of this review is to check for three things, correctness, completeness,

and is it clearly written. If the Critic Agent deems the response to be too vague, incomplete, or even poorly worded, it will rewrite the response.

Input: Draft Response

Output: Final Response

Communication: The Critic agent passes on the final response to the voice agent for audio conversion.

5. Voice Agent

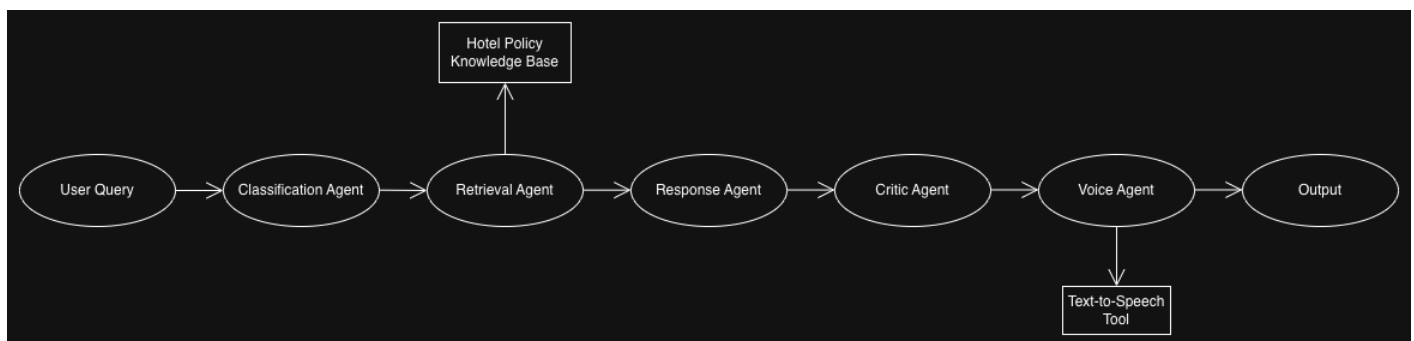
Role: The Voice Agent converts the final text response into an audio response. It reads the final response text and uses a text-to-speech (TTS) tool to generate the audio file.

Input: Final Response Text

Output: Audio Response

Communication: Both the written and audio response are then provided to the user in the output.

Agent Architectural Design

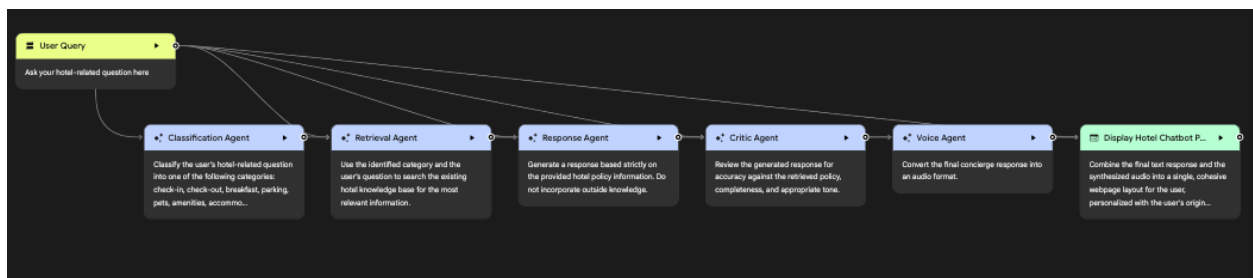


This is the architectural diagram for my multi-agent workflow. The coordination mechanism for this workflow is a fixed pipeline where each agent executes sequentially. The

workflow follows the fixed path Classification Agent→Retrieval Agent→Response Agent→Critic Agent→Voice Agent. The user query passes through the classification agent first where the query is classified into a category. The retrieval agent then uses that classification to search the hotel policy knowledge base and sends the policy information to the response agent. The response agent generates a response based on the policy information. That response is checked by the critic agent and edited where needed. Finally, the written response is sent to the voice agent which calls a TTS tool and converts it to an audio file. The output is then presented to the user.

Updated Code/Implementation

The screenshots below provide a further look into the multi-agent workflow in Google Opal. There are also examples of the user interface. There are five agents in the workflow and each agent is responsible for a separate task. They are connected sequentially in a fixed workflow where each agent passes its output on to the next.



Hotel Chatbot

Draft

Editor

App

Saved

Publish

Share

User Input

Generate

Output

Add Assets

User Query

Ask your hotel-related question here

Classify the user's hotel-related question into one of the following categories: check-in, check-out, breakfast, parking, pets, amenities, etc.

Use the identified category and the user's question to search the existing hotel knowledge base for the most relevant information.

Generate a response based strictly on the provided hotel policy information. Do not incorporate outside knowledge.

Review the generated response for accuracy against the retrieved policy, completeness, and appropriate tone.

Convert the final concierge response into an audio format.

Combine the final text response and the synthesized audio into a single, cohesive message ready for the user, personalized with the user's origin.

Display Hotel Chatbot P...

Edit these steps

Opal can make mistakes, so double-check it

Preview

Console

Stop

Theme

Ask your hotel-related question here

Skip

Type or upload your response.

Opal can make mistakes, so double-check it

Hotel Chatbot

Draft

Editor

App

Saved

Publish

Share

User Input

Generate

Output

Add Assets

User Query

Ask your hotel-related question here

Classify the user's hotel-related question into one of the following categories: check-in, check-out, breakfast, parking, pets, amenities, etc.

Use the identified category and the user's question to search the existing hotel knowledge base for the most relevant information.

Generate a response based strictly on the provided hotel policy information. Do not incorporate outside knowledge.

Review the generated response for accuracy against the retrieved policy, completeness, and appropriate tone.

Convert the final concierge response into an audio format.

Combine the final text response and the synthesized audio into a single, cohesive message ready for the user, personalized with the user's origin.

Display Hotel Chatbot P...

Edit these steps

Opal can make mistakes, so double-check it

Preview

Console

Stop

Theme

Hotel Concierge Chatbot

You: how late can i check out?

Concierge: Our check-out time is 11:00 AM.

0:02 / 0:02

Share output

Download file

